Activity 4.2: Similarity Rules, Right Triangles & Dilations

A

Name: _____ Partner's name: _____ Date: _____

Directions: Answer each question. For #1–3, choose AA, SSS∼, or SAS∼. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

$\angle A \cong \angle D$ and $\angle B \cong \angle E$. Which rule shows $\triangle ABC \sim \triangle DEF$?

Answer: _____ Match? ☐ Yes ☐ No

2

$\triangle ABC$ has sides 3, 4, 5. $\triangle DEF$ has sides 6, 8, 10. Which rule shows they are similar?

Answer: _____ Match? ☐ Yes ☐ No

3

$\dfrac{AB}{DE} = \dfrac{AC}{DF} = 2$ and $\angle A \cong \angle D$. Which rule shows $\triangle ABC \sim \triangle DEF$?

Answer: _____ Match? ☐ Yes ☐ No

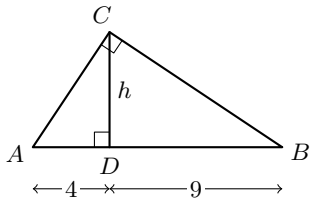
4

Find the geometric mean of 4 and 16.

Answer: _____ Match? ☐ Yes ☐ No

5

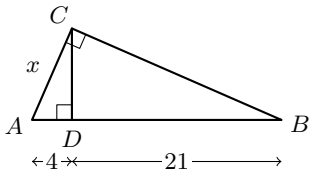
Find the altitude h to the hypotenuse.



Answer: _____ Match? ☐ Yes ☐ No

6

Find x, the length of the leg next to the 4-unit segment.



Answer: _____ Match? ☐ Yes ☐ No

7

A dilation with center $(0, 0)$ and scale factor 3 maps $(2, -1)$ to what point?

Answer: _____ Match? ☐ Yes ☐ No

8

A dilation centered at the origin maps $A(2, 4)$ to $A'(5, 10)$. Find the scale factor.

Answer: _____ Match? ☐ Yes ☐ No

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B

Name: _____ Partner's name: _____ Date: _____

Directions: Answer each question. For #1–3, choose AA, SSS∼, or SAS∼. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

$\angle P \cong \angle X$ and $\angle Q \cong \angle Y$. Which rule shows $\triangle PQR \sim \triangle XYZ$?

Answer: _____ Match? ☐ Yes ☐ No

2

$\triangle PQR$ has sides 5, 12, 13. $\triangle XYZ$ has sides 10, 24, 26. Which rule shows they are similar?

Answer: _____ Match? ☐ Yes ☐ No

3

$\dfrac{PQ}{XY} = \dfrac{PR}{XZ} = 3$ and $\angle P \cong \angle X$. Which rule shows $\triangle PQR \sim \triangle XYZ$?

Answer: _____ Match? ☐ Yes ☐ No

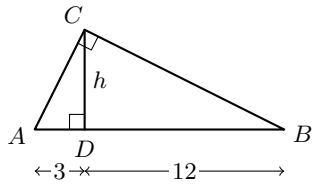
4

Find the geometric mean of 2 and 32.

Answer: _____ Match? ☐ Yes ☐ No

5

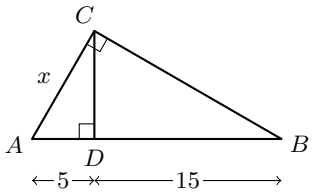
Find the altitude h to the hypotenuse.



Answer: _____ Match? ☐ Yes ☐ No

6

Find x, the length of the leg next to the 5-unit segment.



Answer: _____ Match? ☐ Yes ☐ No

7

A dilation with center $(0, 0)$ and scale factor $\dfrac{3}{2}$ maps $(4, -2)$ to what point?

Answer: _____ Match? ☐ Yes ☐ No

8

A dilation centered at the origin maps $B(4, 6)$ to $B'(10, 15)$. Find the scale factor.

Answer: _____ Match? ☐ Yes ☐ No